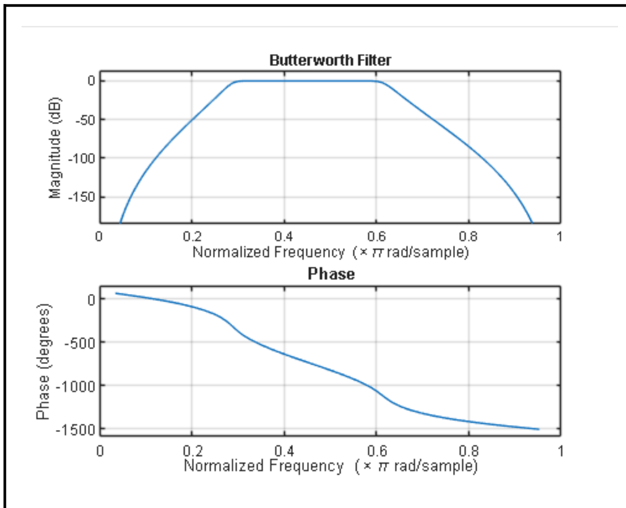
3 .Performance Analysis of IIR Filters

3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

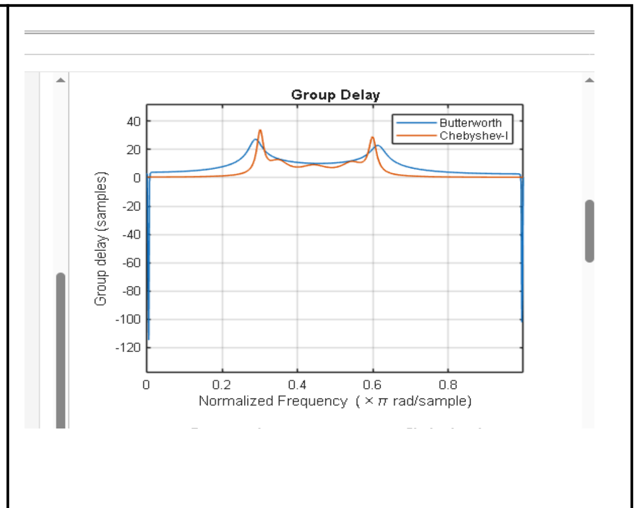
Program

```
clc; clear; close all;
% Specifications
Fs=2000; Rp=1; Rs=40;
Wp=[300 600]/(Fs/2);
Ws=[200 700]/(Fs/2);
% Butterworth Filter
[n1,W1]=buttord(Wp,Ws,Rp,Rs);
[b1,a1]=butter(n1,W1,'bandpass');
% Chebyshev Type-I Filter
[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2]=cheby1(n2,Rp,W2,'bandpass');
% Magnitude Response
figure;
freqz(b1,a1); title('Butterworth Filter');
figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');
% Group Delay
figure;
grpdelay(b1,a1); hold on;
grpdelay(b2,a2);
legend('Butterworth','Chebyshev-I');
% Pole-Zero Plot
figure;
subplot(121),zplane(b1,a1),title('Butterworth')
subplot(122),zplane(b2,a2),title('Chebyshev-I')
% Impulse Response
figure;
subplot(211),impz(b1,a1),title('Butterworth Impulse')
subplot(212),impz(b2,a2),title('Chebyshev Impulse')
% Pole Magnitudefigure;
subplot(121),stem(abs(roots(a1)),'filled')
title('Butterworth Pole Magnitude')
subplot(122),stem(abs(roots(a2)),'filled')
title('Chebyshev Pole Magnitude')
disp(['Butterworth Order = ',num2str(n1)])
disp(['Chebyshev Order = ',num2str(n2)])
```

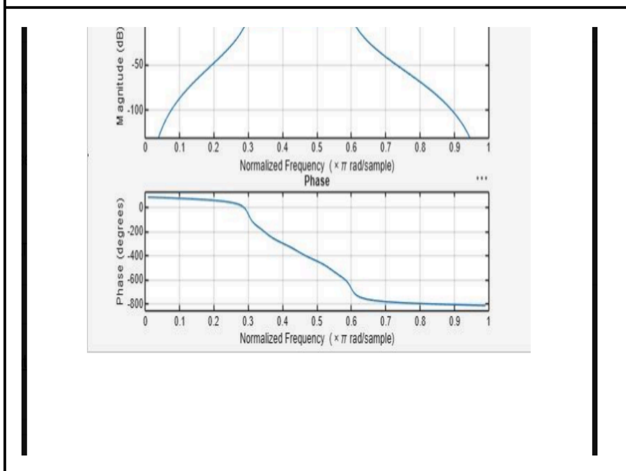
Output :



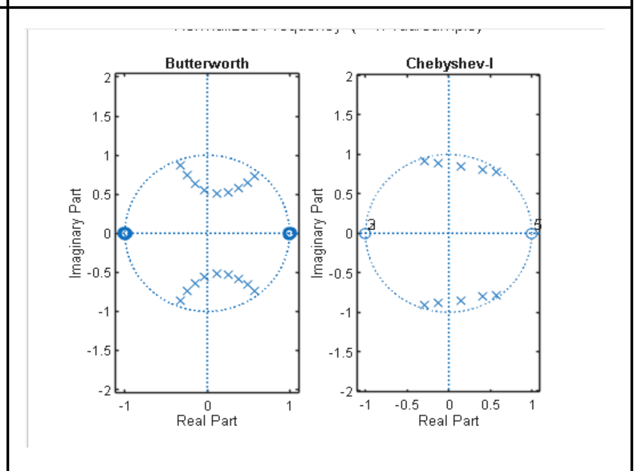
Butterworth Filter
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)



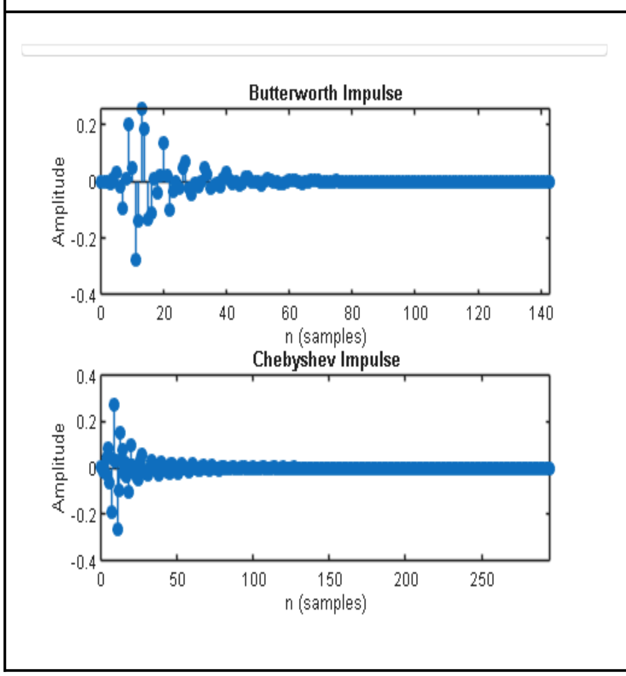
Group Delay
Butterworth
Chebyshev-I
Group delay (samples)
Normalized Frequency (× π rad/sample)



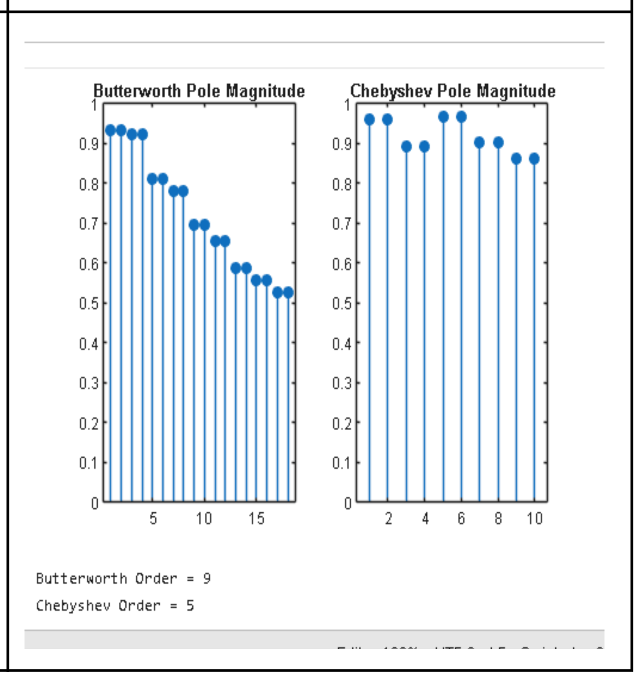
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)



Butterworth
Chebyshev-I
Imaginary Part
Real Part



Butterworth Impulse
Amplitude
n (samples)
Chebyshev Impulse
Amplitude
n (samples)



Butterworth Pole Magnitude
Chebyshev Pole Magnitude
Butterworth Order = 9
Chebyshev Order = 5